Arista Access Node JAY-ACC-47 Optical Transport Failure Alarm ALARM_4715 Resolution SOP

Node: JAY-ACC-47 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_4715

Description

The specified failure domain of Optical Transport Failure on the Arista Access Node JAY-ACC-47 is attributed to hardware degradation or environmental effects on the SFP/XFP modules or the optical fibers. The Arista Access Node's role is to provide Layer 2 connectivity to the attached devices, relying on the optical transport to facilitate data transmission. This SOP focuses on resolving the Alarm ALARM_4715, characterized as a Minor severity, which is likely triggered by a degradation in the SFP/XFP module's receive or transmit power, a physical cut in the fiber, or other environmental factors affecting the optical link.

Pre-requisite Checks

- Verify the site location and ensure the Jayanagar Metro Site is accessible.
- Confirm the redundant link status on the Arista Access Node JAY-ACC-47.
- Ensure dispatch of remote hands to the site, if required, for physical intervention.

Potential Root Causes

Cause	Probability
SFP/XFP module degradation or failure	High
Physical cut in the fiber optic link	Medium
Environmental effects on the optical link (temperature, humidity, etc.)	Low

Diagnostic Commands & Expected Outputs

```
Command: show interfaces status | include JAY-ACC-47, Expected Output: JAY-ACC-47: 10G-ER: not present, Description: This command checks the status of the interface on the Arista Access Node JAY-ACC-47 and determines if the SFP/XFP module is present and functioning.

Command: show optics, Expected Output: SFP-10G-SR: TX power 0.5 dBm, RX power -3.2 dBm, Description: This command provides information about the optical module's transmit and receive power levels, which can indicate potential issues with the SFP/XFP module.

Command: show environment, Expected Output: Temperature: 25°C, Humidity: 60%, Description: This command checks the environmental conditions at the site, which can impact the optical link's performance.
```

Resolution Steps

Step 1: Step: Remove and Inspect the SFP/XFP module, Command: no interfaces 1/1/1, Description: Remove the SFP/XFP module from the Arista Access Node JAY-ACC-47 and inspect it for any visible signs of damage or degradation.

Step 2: Step: Replace the SFP/XFP module, Command: interface 1/1/1, Description: Replace the SFP/XFP module with a new one and secure it properly in the Arista Access Node JAY-ACC-47.

Step 3: Step: Verify the optical link, Command: show interfaces status | include JAY-ACC-47, Description: Verify that the SFP/XFP module is detected and the optical link is functioning properly after replacement.

Post-Check Verification

```
Command: show interfaces status | include JAY-ACC-47, Expected Output:
JAY-ACC-47: 10G-ER: present, Description: Verify that the SFP/XFP
module is detected and the optical link is functioning properly.

Command: show optics, Expected Output: SFP-10G-SR: TX power 0.5 dBm,
RX power -3.2 dBm, Description: Verify that the optical module's
transmit and receive power levels are within the expected range.
```

Roll-back Procedures

- Step: Reinstall the original SFP/XFP module, Command: interfaces 1/1/1, Description: Reinstall the original SFP/XFP module to revert to the previous state.

- Step: Verify the original configuration, Command: show run, Description: Verify that the original configuration is restored on the Arista Access Node JAY-ACC-47.